

Alexander Soto

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EDUCATION

Stanford University Stanford, CA
B.S. Electrical Engineering; Concentration: Hardware & Software; GPA: 3.91 Sep. 2022 - June 2026

Stanford University Stanford, CA
M.S. Electrical Engineering Sep. 2026 - June 2027

PROFESSIONAL EXPERIENCE

Stanford University - Electrical Engineering Research Intern | Stanford, CA June 2025 – Present

- **Training deep learning models** to improve the speed and accuracy of PET (Positron Emission Tomography) image reconstruction.
- Generating simulated PET imaging data using the SimSET Monte Carlo toolkit.
- Enhancing and documenting the **Deep Learning Filtered Backprojection pipeline** for future researchers.
- Learning the fundamentals of PET imaging, including physics and image reconstruction.

Tyson Foods - Robotics/Machine Learning Intern | Springdale, AR June 2024 – Sept. 2024

- Explored Universal Manipulation Interface (UMI), a system used to **transfer human skills to robots**, to further automate Tyson's plants.
- Assembled a pair of UMI grippers and collected robust training demonstrations.
- **Trained diffusion policies** modeling various human tasks.
- Learned the fundamentals of state-of-the-art **behavior cloning**.

University of Arkansas - Biomedical Engineering Research Intern | Fayetteville, AR June 2023 – Aug. 2023

- **Trained a convolutional neural network** in MATLAB to segment cells in NADH/FAD autofluorescence images to facilitate metabolic analysis.
- Fit a model that performed with **86% accuracy** on out of sample test images.
- Presented results at the national Biomedical Engineering Society conference.

PROJECTS

MIPS Processor

- **Implemented a 5-stage pipelined MIPS processor in Verilog**, incorporating full hazard detection, data forwarding, and branch handling.
- Validated the functional performance of the processor by **executing an optimized image convolution program** on the architecture.

FPGA Music Player

- **Built a real-time FPGA music player** with chords, dynamics, and echo effects.
- **Implemented an interactive UI** featuring effect-toggle buttons and a live waveform visualization.

RELEVANT COURSES

ME 210: Introduction to Mechatronics

EE 114: Fundamentals of Analog Integrated Circuit Design

EE101A: Circuits I

EE101B: Circuits II

EE180: Digital Systems Architecture

EE108: Digital System Design

EE102A: Signals and Systems I

EE104: Introduction to Machine Learning

CS107: Computer Organization and Systems

CS106B: Programming Abstractions